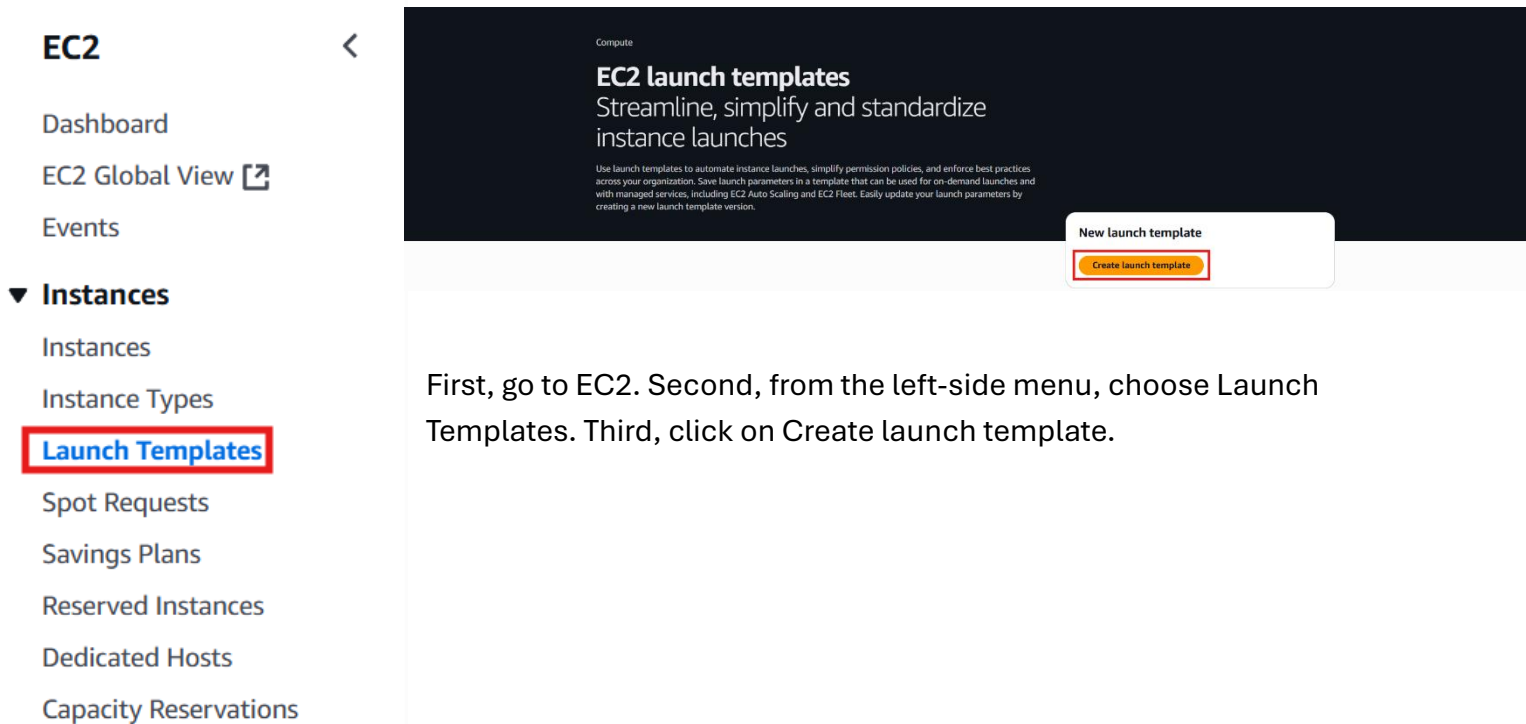
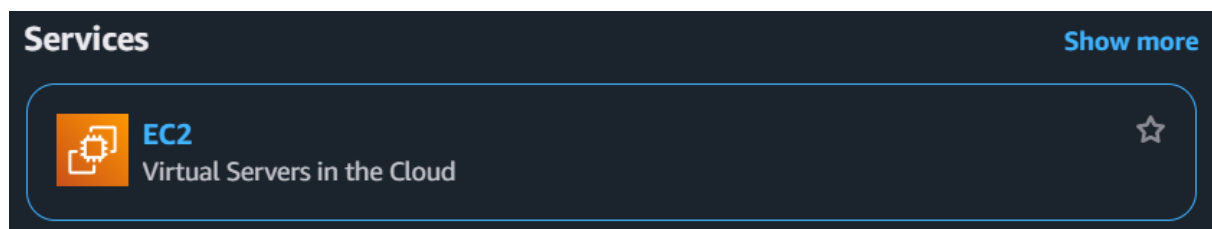


How to Create Auto Scaling groups

Requirement:

1. 3 subnets (2 Public and 1 private).
2. Load balancer
3. Security Group
4. Key pair
5. **SNS topic with subscription on email or anything**

How to Create a Launch Template



Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

MyTemplate

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Choose a name and write a meaningful description.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Don't include in launch template	Amazon Linux aws	macOS Mac	Ubuntu ubuntu	Windows Microsoft	Red Hat Red Hat	SUSE Linux SUSE	Debian debian
----------------------------------	---------------------	--------------	------------------	----------------------	--------------------	--------------------	------------------

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-084568db4383264d4 (64-bit (x86)) / ami-0c4e709339fa8521a (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

64-bit (x86)

AMI ID

ami-084568db4383264d4

Publish Date

2025-03-05

Username

ubuntu

Verified provider

For the **Application and OS Images (Amazon Machine Image)**, choose Quick start, then **Ubuntu**.

▼ Instance type [Info](#) [Get advice](#)

Advanced

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

For the instance type, choose **t2.micro**

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

Don't include in launch template ▼ [Create new key pair](#)

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template ▼ [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups [Info](#)

Select security groups ▼ [Compare security group rules](#)

▶ Advanced network configuration

For the Key pair, **choose your key pair**. In the subnet, **leave it. Don't include it in the Launch template**. In Firewall (security groups), select the **security groups** you created before.

Click on Advanced network configuration, then click on Add network interface

▼ Advanced network configuration

Network interface 1

Device index [Info](#)

0

Network interface [Info](#)

New interface ▼

Description [Info](#)

Subnet [Info](#)

Don't include in launch template

Security groups [Info](#)

Select security groups ▼ [Show all selected \(1\)](#)

Auto-assign public IP [Info](#)

Enable ▼

Primary IP [Info](#)

123.123.123.1

Secondary IP [Info](#)

Don't include in launch template ▼

IPv6 IPs [Info](#)

Don't include in launch template ▼

IPv4 Prefixes [Info](#)

Don't include in launch template ▼

The selected instance type does not support IPv4 prefixes.

IPv6 Prefixes [Info](#)

Don't include in launch template ▼

The selected instance type does not support IPv6 prefixes.

Assign Primary IPv6 IP [Info](#)

Don't include in launch template ▼

Delete on termination [Info](#)

Don't include in launch template ▼

Interface type [Info](#)

Don't include in launch template ▼

Network card index [Info](#)

Don't include in launch template ▼

The selected instance type does not support multiple network cards.

ENA Express [Info](#)

Don't include in launch template ▼

The selected instance type does not support ENA Express.

ENA Express UDP [Info](#)

Don't include in launch template ▼

The selected instance type does not support ENA Express.

Idle connection tracking timeout [Info](#)

☐ Enable

Idle connection tracking timeout is only supported on Nitro instances.

[Add network interface](#)

Click on Auto-assign public key, choose “Enable” so that the **Auto Scaling group assigns a public IP address**.

▼ Storage (volumes) [Info](#)

EBS Volumes [Hide details](#)

▶ Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3)
AMI Volumes are not included in the template unless modified

① Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

▼ Resource tags [Info](#)

No resource tags are currently included in this template. Add a resource tag to include it in the launch template.

Add new tag

You can add up to 50 more tags.

▶ Advanced details [Info](#)

Scroll down and **click on advanced details.**

Metadata version [Info](#)

V2 only (token required)

⚠ For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit [Info](#)

2

Allow tags in metadata [Info](#)

Don't include in launch template

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
sudo apt -y update
sudo apt install -y apache2
sudo systemctl enable apache2
sudo systemctl start apache2

echo "<h1>Hello From $(hostname -f)</h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Scroll down to the end. You will find **user data**. Add these commands. These commands will update the system and install Apache, and in the index.html, it will print Hello from and the username of the machine, so that you know what machine the load balancer will be using.

▼ Summary

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-084568db4383264d4

Virtual server type (instance type)

t2.micro

Firewall (security group)

Hussain-SG

Storage (volumes)

1 volume(s) - 8 GiB

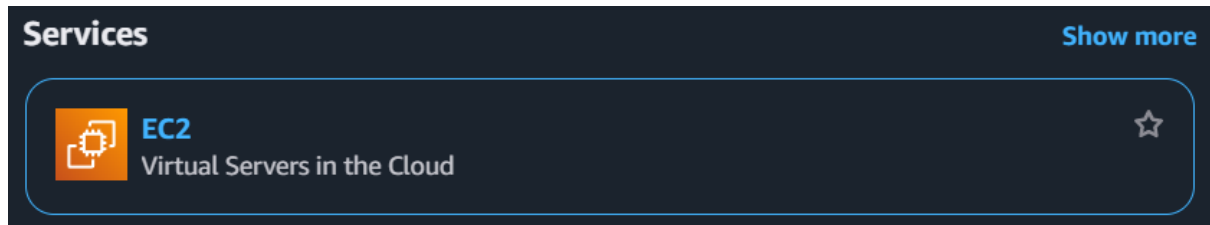
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#)

[Create launch template](#)

Click on **Create launch template**.

Create Auto scaling group



EC2

Dashboard

EC2 Global View 

Events

► Instances

► Images

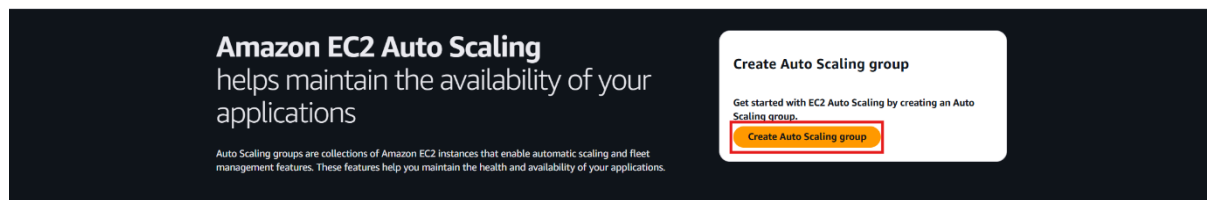
► Elastic Block Store

► Network & Security

► Load Balancing

▼ Auto Scaling

Auto Scaling Groups



In EC2, in the left-side menu under Auto Scaling, click on Auto Scaling Groups, then click **Create Auto Scaling Group**.

Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
 
[Create a launch template](#)

[Cancel](#) [Next](#)

Choose a name for your Auto scaling group, then **choose your template** that you created before, and then click **Next**.

Instance type requirements [Info](#) [Override launch template](#)

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
Hussain-TEP lt-00db2581e2ede0a2d	Default	website

Instance type
t2.micro

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
 
172.31.0.0/16 Default
[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
 
[Create a subnet](#)

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

First, **choose your VPC**. Second, **choose your subnets at least 2 subnets in different availability zones**. Third, click **Next**.

Integrate with other services - optional [Info](#)

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer

Choose from your existing load balancers.

☐ Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ Choose from your load balancer target groups

This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

In Load balancing, choose **Attach to an existing load balancer**, then **choose your target group** under **Existing load balancer target groups**.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - optional [Info](#)

☐ Turn on Elastic Load Balancing health checks

Recommended

Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ Turn on VPC Lattice health checks

VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks

EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300

seconds

[Cancel](#)

[Skip to review](#)

[Previous](#)

Next

Scroll down and click **Next**, and leave everything as the default option.

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

4

Equal or greater than desired capacity

In the Desired capacity, **choose how many EC2 instances will be made**, the **minimum number of instances that will be running**, and the **maximum number of instances that will be running at the same time**.

Scaling [Info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Equal or less than desired capacity

Max desired capacity

Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ **No scaling policies**
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ **Target tracking scaling policy**
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Metric type [Info](#)
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Target group

Target value

Instance warmup [Info](#)
 seconds

☐ Disable scale in to create only a scale-out policy

For Automatic scaling, choose Target tracking scaling policy because you need the Auto Scaling to scale in certain conditions, and that is when the load balancer receives a number of requests, the Auto Scaling will work without the need for you to enable it. Second, choose your target group and the value that, when it hits, the Auto scaling will start to scale EC2 instances.

Additional capacity settings

Capacity Reservation preference [Info](#)

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

- ☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.
- ☐ **None**
Instances will not be launched into a Capacity Reservation.
- ☐ **Capacity Reservations only**
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
- ☐ **Capacity Reservations first**
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

- ☐ Enable instance scale-in protection

Monitoring [Info](#)

- ☐ Enable group metrics collection within CloudWatch

Default instance warmup [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

- ☐ Enable default instance warmup

[Cancel](#)

[Skip to review](#)

[Previous](#)

[Next](#)

Scroll down, leave the rest as default, and click Next.

Add notifications - optional [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

[Add notification](#)

[Cancel](#)

[Skip to review](#)

[Previous](#)

[Next](#)

Click on **Add notification**

From this point on, double-check that you have the **SNS topic** ready since we will now add the notifications for when the auto-scaling is active and when creating VMs are created.

Add notifications - *optional* [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

▼ Notification 1 Remove

SNS Topic
Choose an SNS topic to use to send notifications

Create a topic

Event types
Notify subscribers whenever instances
☒ Launch
☒ Terminate
☒ Fail to launch
☒ Fail to terminate

Add notification

Cancel Skip to review Previous Next

Choose your SNS Topic, then choose next.

Add tags - *optional* [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

1 You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group. ×

Tags (0)
Add tag
50 remaining

Cancel Previous Next

Choose **Next**.

Step 6: Add tags Edit

Tags (0)

Key	Value	Tag new instances
No tags		

Preview code Cancel Previous Create Auto Scaling group

Scroll down and click on **Create Auto Scaling group**.

Auto Scaling groups (1) [Info](#)

Search your Auto Scaling groups

Last updated less than a minute ago Launch configurations Launch templates Actions Create Auto Scaling group

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
☐	Hussain-AS	Hussain-TEP Version Default	0	Updating capacity...	2	1	4	us-east-1b, us-east-1c

Wait for a couple of minutes, and you will receive an email for the EC2 instance that has been created.

Instances info

Find Instance by attribute or tag (case-sensitive) All states

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
No matching instances found										

You can see **before** the Auto Scaling runs.

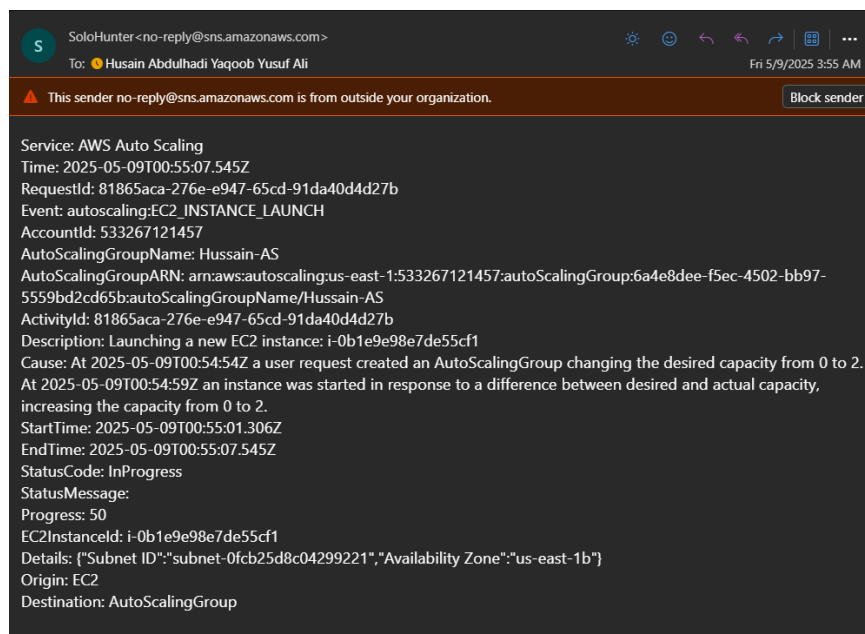
Instances (2) info

Find Instance by attribute or tag (case-sensitive) All states

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
	i-0abb71fd03cad7d1	Running	t2.micro	Initializing	View alarms +	us-east-1c	-	3.85.48.243	-	-
	i-0b1e9e98e7de55cf1	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	54.196.173.223	-	-

After the Auto scaling runs, you will see 2 EC2 instances have been deployed.



This is a sample of what you will get from the Auto scaling through **Email**.

Hello From ip-192-168-2-236

Hello From ip-192-168-3-153

As you can see, the load **balancer works perfectly with auto scaling**.